

**WELCOME
TO
Lecture session on
Wireless Communication systems
(BEC703)**

By

**Dr. BHASKAR S
PROFESSOR IN ECE DEPARTMENT
SJCIT, CHICKBALLAPUR**

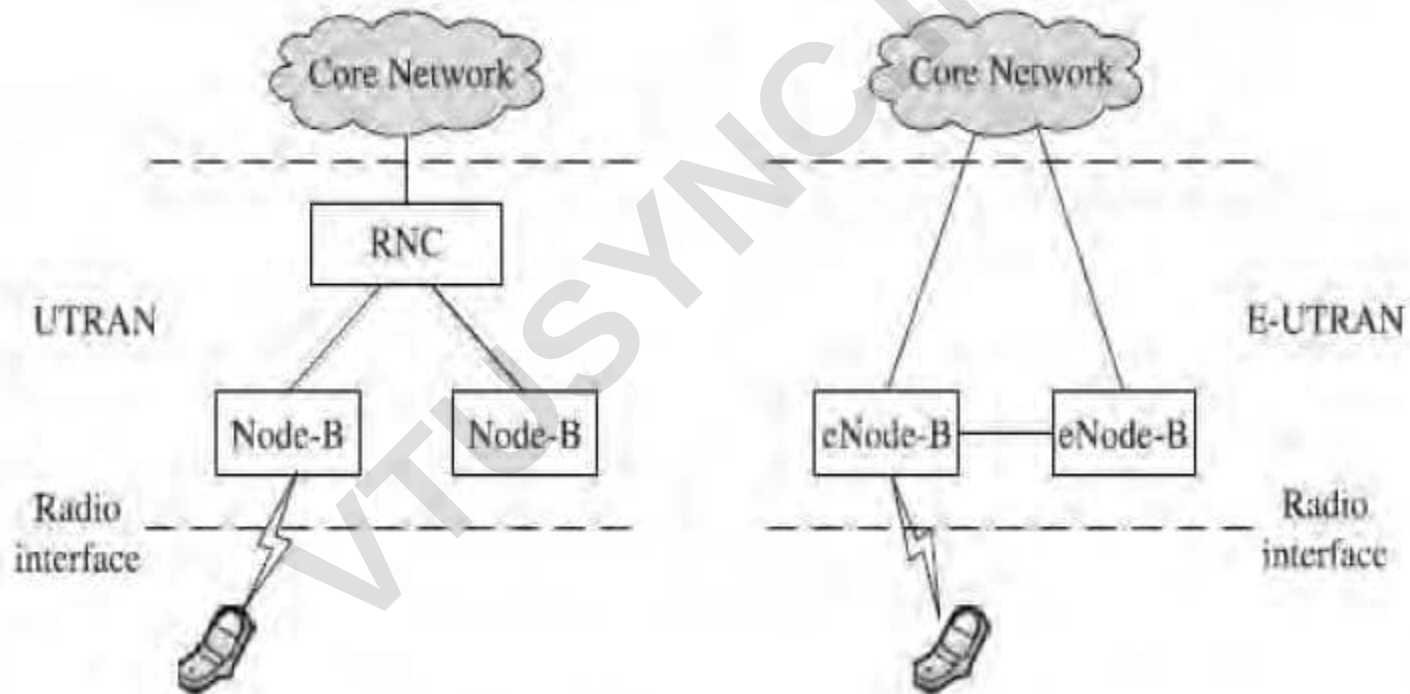
Overview and Channel Structure of LTE

Introduction to LTE :

- LTE is the next step in the evolution of mobile cellular systems.
- Unlike 2G and 3G cellular systems that were designed mainly with **voice services in mind**,
- LTE was designed primarily for **high-speed data services**, which is why LTE is a packet-switched network from end to end and has no support for circuit-switched services.
- However, the low latency of LTE and its sophisticated **quality of service (QoS)** architecture allow a network to emulate a circuit-switched connection on top of the packet-switched framework of LTE.

Radio interface architectures of UTRAN and E-UTRAN.

RNC- Radio N/W controller, UTRAN-UMTS Terrestrial Radio Access N/W,



Design Principles:

Network Architecture:

- Unlike 3G networks, LTE was designed to support packet-switched traffic with support for various QoS classes of services.
- LTE is different in the sense that it is a clean slate design and supports packet switching for high data rate services from the start.
- The LTE radio access network, E-UTRAN, was designed to have the minimum number of interfaces (i.e., the minimum number of network elements) while still being able to provide efficient packet-switched transport for traffic belonging to all the QoS classes such as conversational, streaming, real-time.

Data Rate and Latency:

- The design target for downlink and uplink peak data rates for LTE are 100 Mbps and 50 Mbps, respectively, when operating at the 20MHz.
- The user-plane latency is defined in terms of the time it takes to transmit a small IP packet from the UE to the edge node of the radio access network or vice versa measured on the IP layer.
- The target for one-way latency in the user plane is 5 ms in an unloaded network, that is, if only a single UE is present in the cell. For the control-plane latency, the transition time for active state is less than 100 ms, while the transition time between a comped state and an active state should be less than 50 MS.

Spectrum Efficiency

- The average downlink user data rate and spectrum efficiency target is three to four times that of the baseline HSDPA network. Similarly, in uplink the average user data rate and spectrum efficiency.
- Mobility The mobility requirement for LTE is to be able to support hand off/ mobility at different terminal speeds.
- Maximum performance is expected for the lowest terminal speeds of 0 to 15 km/hr, with minor degradation in performance at higher mobile speeds up to 120 km/hr.
- LTE is also expected to be able to sustain a connection for terminal speeds up to 350 km/hr but with significant degradation in the system performance.
- Coverage for the cell coverage, the above performance targets should be met up to 5 km.

Radio Resource Management:

➤ The radio resource management requirements over various spots such as enhanced support for end-to-end QoS, efficient support for transmission of higher layers, and support for load sharing/balancing and policy management enforcement across different radio access technologies.

Flexibility of Spectrum and Deployment:

➤ In order to become a truly global standard, LTE is designed to be operable under a wide variety of spectrum, including its ability to coexist and share spectrum with existing 3G technologies.

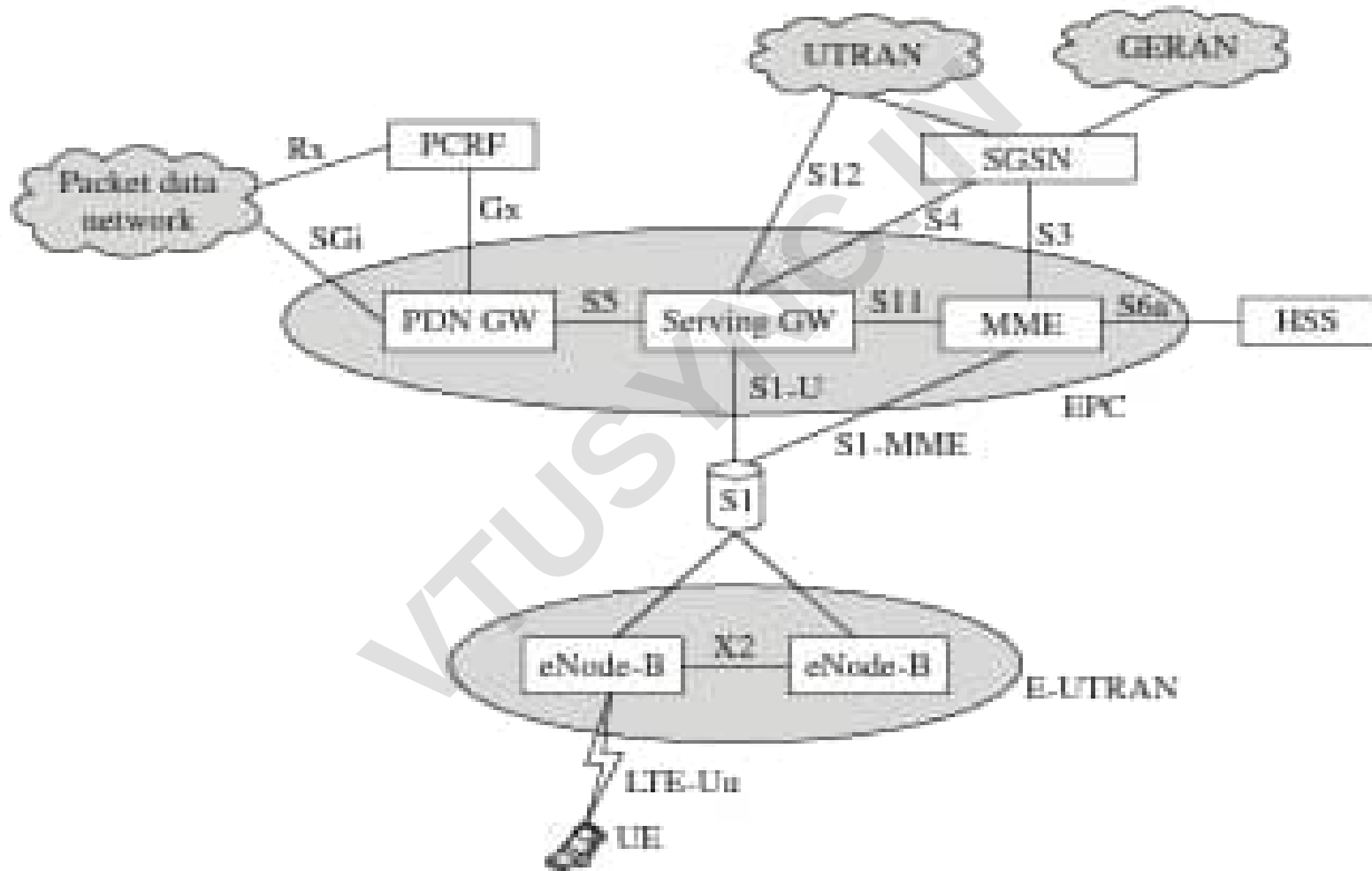
➤ Service providers in different geographical regions often have different spectrums in terms of the carrier frequency and total available bandwidth, which is why LTE designed to have a scalable bandwidth from 1.4MHz to 20MHz.

➤ In order to accommodate flexible duplexing options, LTE designed to operate in both frequency division duplex (FDD) and time division duplex (TDD) modes.

Interoperability with 3G and 2G Networks:

➤ Multimode LTE terminals, which support UTRAN and/or GERAN(GSM/EDGE Radio Access Network) operation, should be able to support measurement of, and handover from into, both 3GPP(Partnership Project) UTRAN und 3GPP GERAN systems with acceptable terminal complexity and network performance.

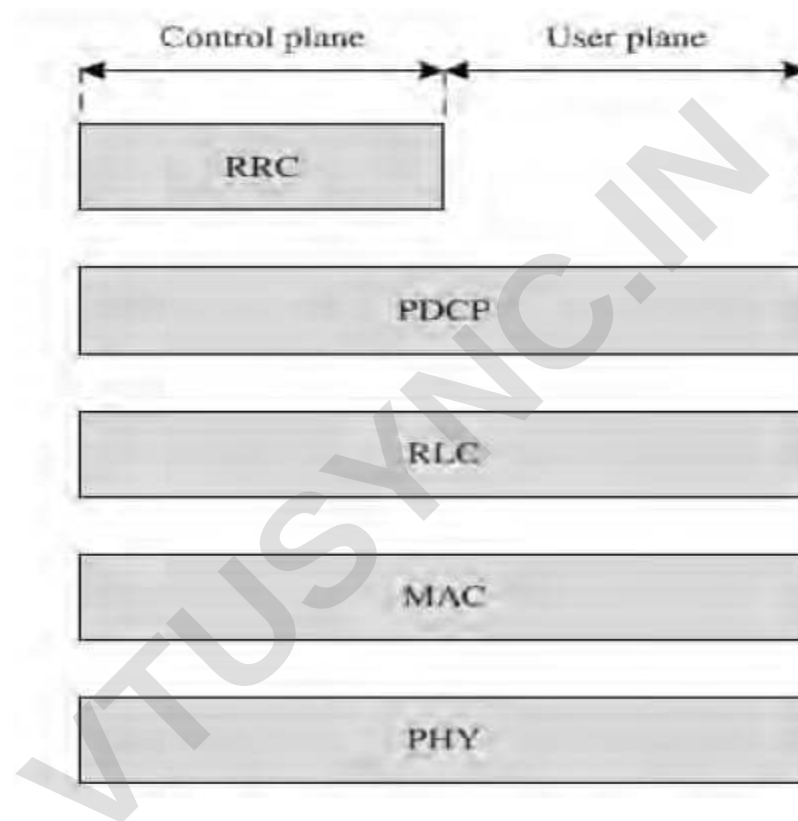
Network Architecture:



LTE end to end network architecture

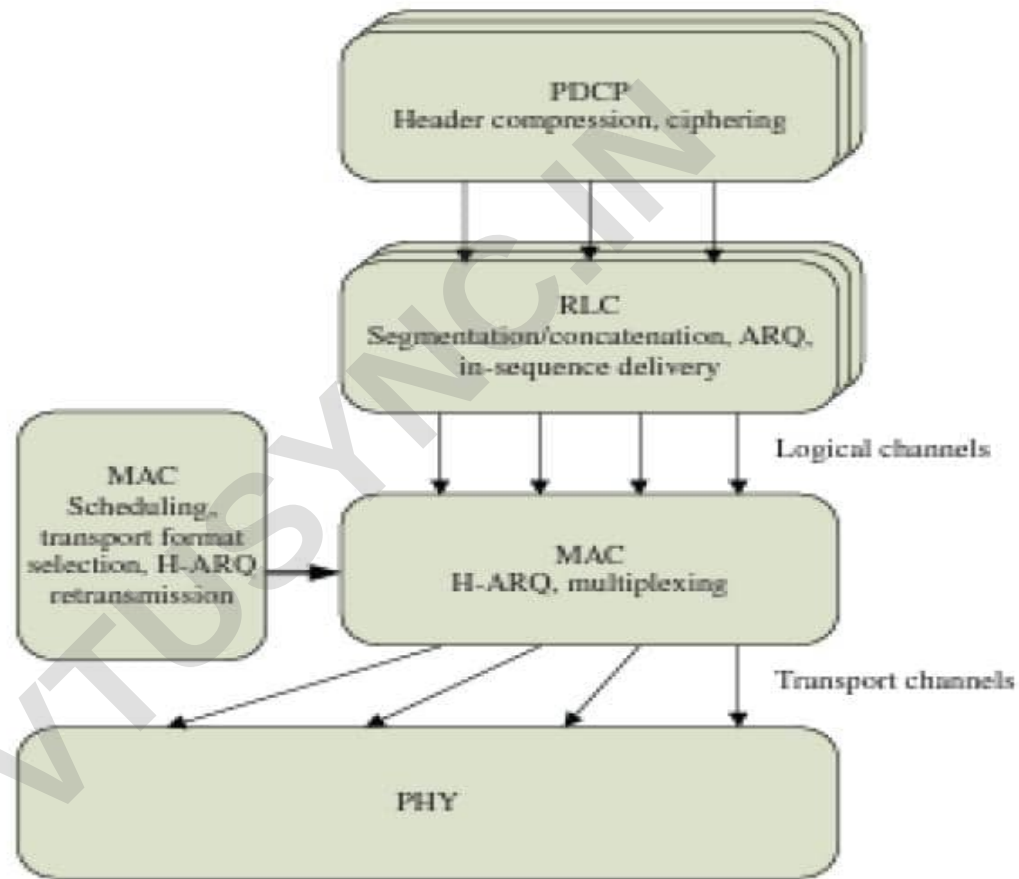
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Radio Interface Protocols:



The LTE radio interface protocol stack

Hierarchical Channel Structure of LTE

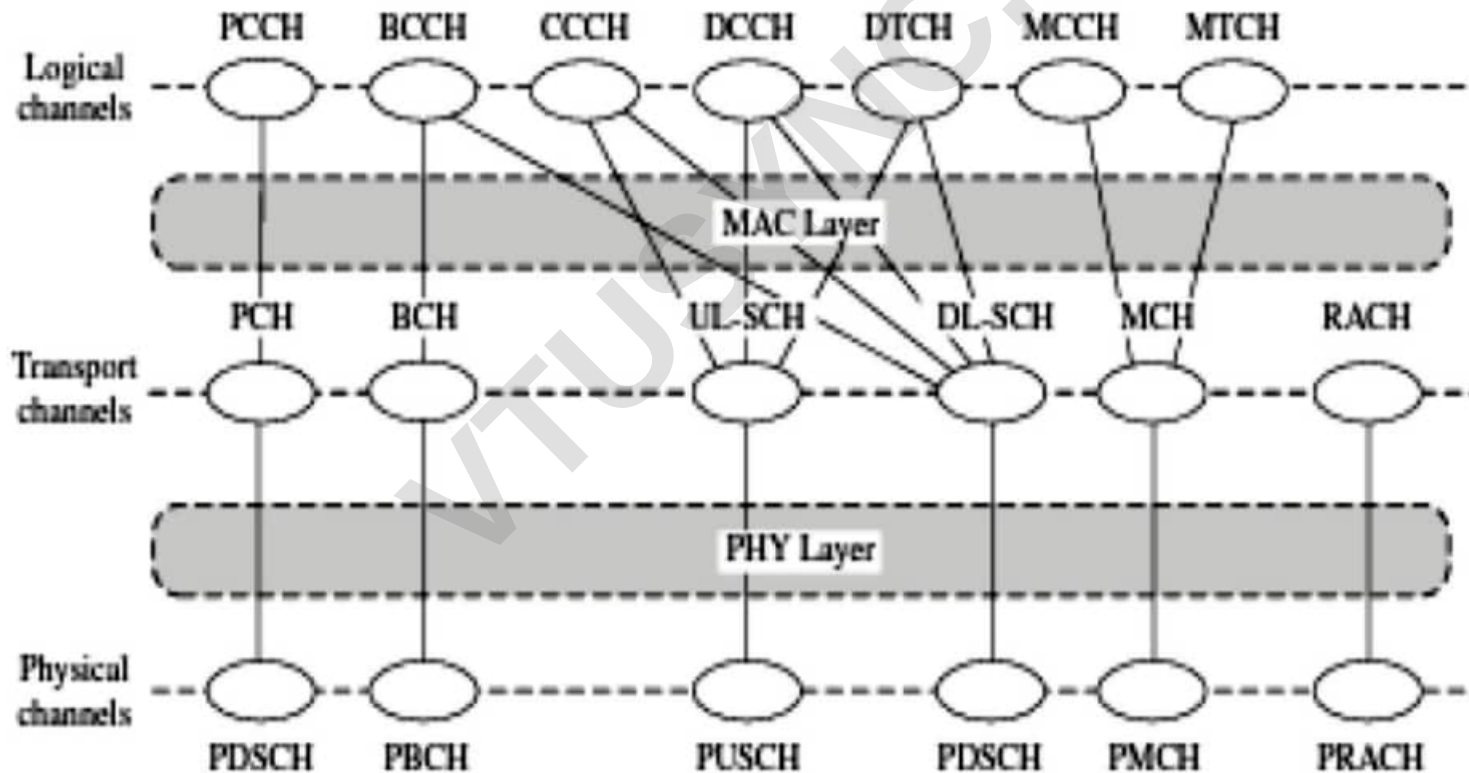


The radio interface protocol architecture and the SAPs b/w different layers

Hierarchical Channel Structure of LTE

3 types of Channels:

1. Logical channels
2. Transport channel
3. Physical channel.



Logical channels

Broadcast control channel(BCH): broadcast system control information to the mobile terminals in the cell, including downlink system bandwidth, antenna configuration, and reference signal power.

Multicast control channel: channel A point-to-multipoint downlink channel used for transmitting control information to UEs in the cell. It is only used by UES that receive multicast/'broadcast services.

Paging control channel: A downlink channel that transfers paging information to registered UEs in the cell, for example, in case of a mobile-terminated communication session.

Common control channel(COCH): A bi-directional channel for transmitting control information between the network and UEs when no RRC connection is available, implying the UE is **not attached to the network such as in the idle state**. Most commonly the COCH is used during the random access procedure.

Dedicated control channel(DCCH): A point-to-point, bi-directional channel that transmitted dedicated control information between a UE and the network. This channel is used when the RRC connection is available, that is, the **UE is attached to the network**.

Dedicated traffic channel(DTCH): A point-to-point, **bi-directional channel** used between a given UE and the network. It can exist in both uplink and downlink

Multicast traffic channel(MTCH): A **unidirectional** point-to-multipoint data channel that transmits traffic data from the network to UEs. It is associated with the multicast/broadcast service.

Transport channel

(a) Downlink transport channel

1. Downlink shared channel
2. Broadcast channel
3. Multicast channel
4. Paging channel

(b) Uplink transport channel

1. Uplink shared channel
2. Random access channel

Downlink Shared Channel (DL-SCH):

- Transmitting the downlink data, including both control and traffic data, and thus it is associated with both logical control and logical traffic channels.
- able to **maximize the throughput** by allocating the resources to the optimum UEs.

Broadcast Channel (BCH): is used to broadcast **system information** over the entire coverage area of the cell. It has a fixed transport format defined by the specifications.

Multicast Channel (MCH): which transmits the same information on the same radio resource from multiple synchronized base stations to multiple UES.

Paging Channel (PCH): It is mapped to dynamically allocated physical resources, and is required for **broadcast over the entire cell** coverage area.

Uplink Transport Channels

Uplink Shared Channel (UL-SCH): The uplink counterpart of the DL-SCH. It can be associated to CCCH, DOCH, and DTCH logical channels. It supports H-ARQ, dynamic link adaptation, and dynamic and semi-persistent resource allocation.

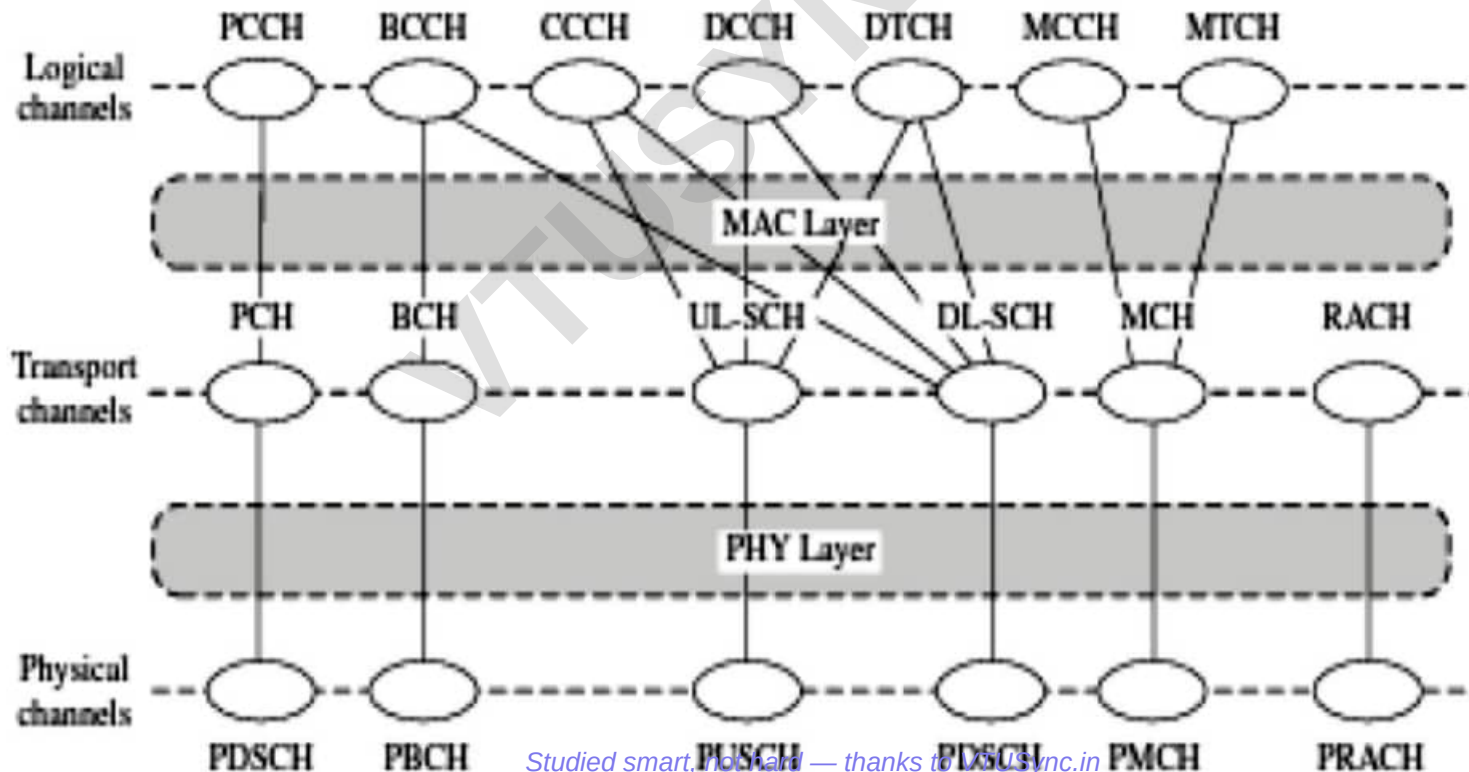
Random Access Channel (RACH): Mobile originating call and data transmission. There will be a small amount of data loss initially.

Downlink Control Information (DCI):

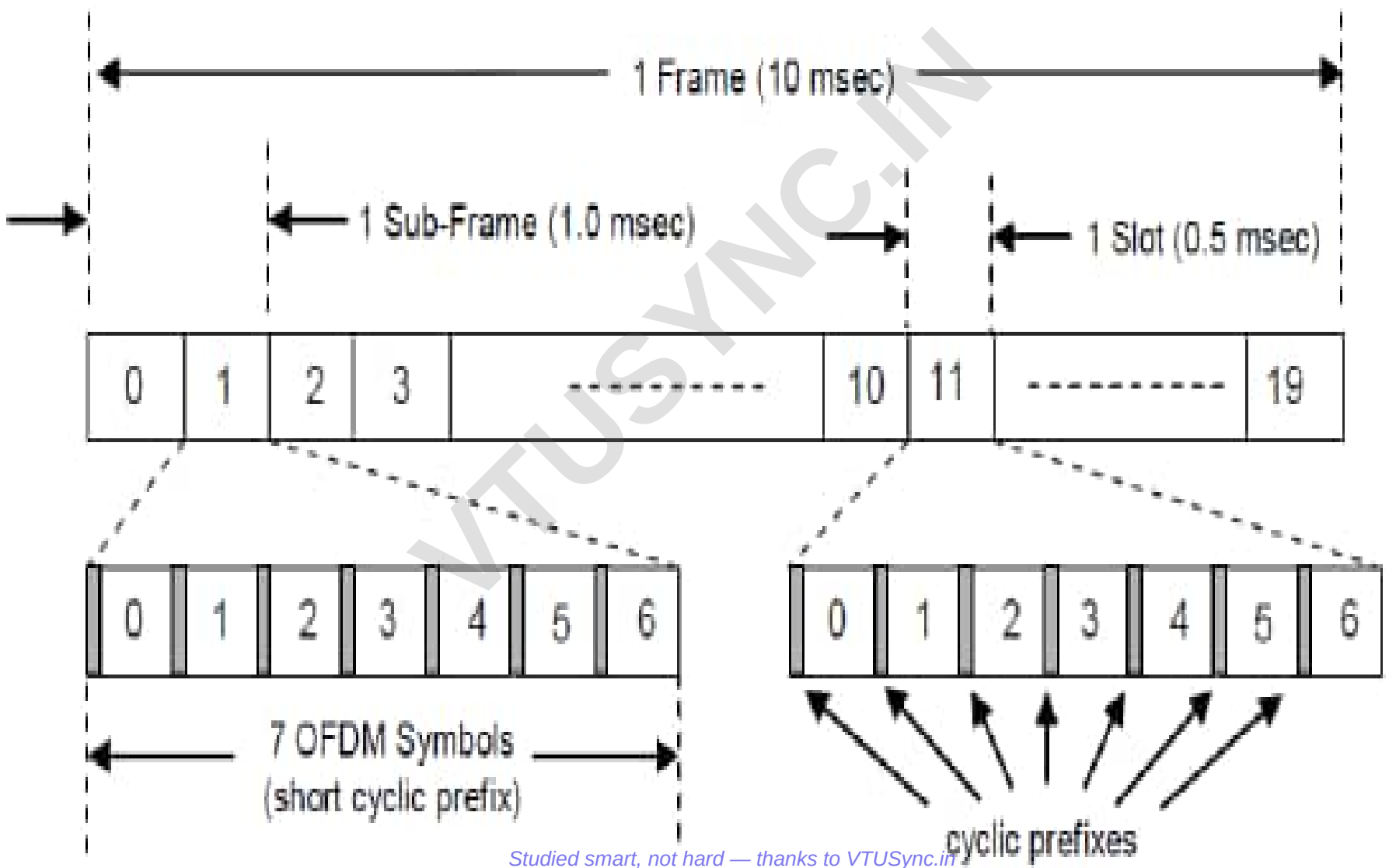
- It carries information related to downlink' uplink scheduling assignment.
- Modulation and coding scheme.
- Transmit Power Control (TPC) command, and is sent over the Physical Downlink Control Channel (PDCCH).

Channel Mapping :

- we see that there exists a good correlation based on the purpose and the content between channels in different layers.
- This requires a mapping between the logical channels and transport channels



Frame Structure: In frame structure type 1. at uplink radio frame consists of 20 slots of 0.5 ms each, and one sub frame consists of two slots



One radio frame, $T_f = 10$ ms

One subframe, 1 ms (a TT1)

One slot, $T_{slot} = 0.5$ ms



OFDM Symbol

$66.7 \mu s$

Normal CP

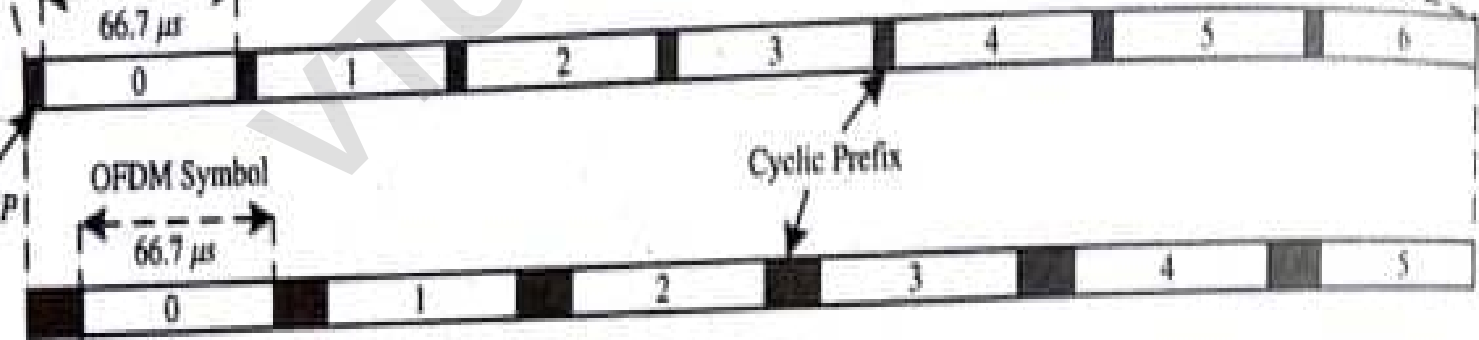
T_{CP}

OFDM Symbol

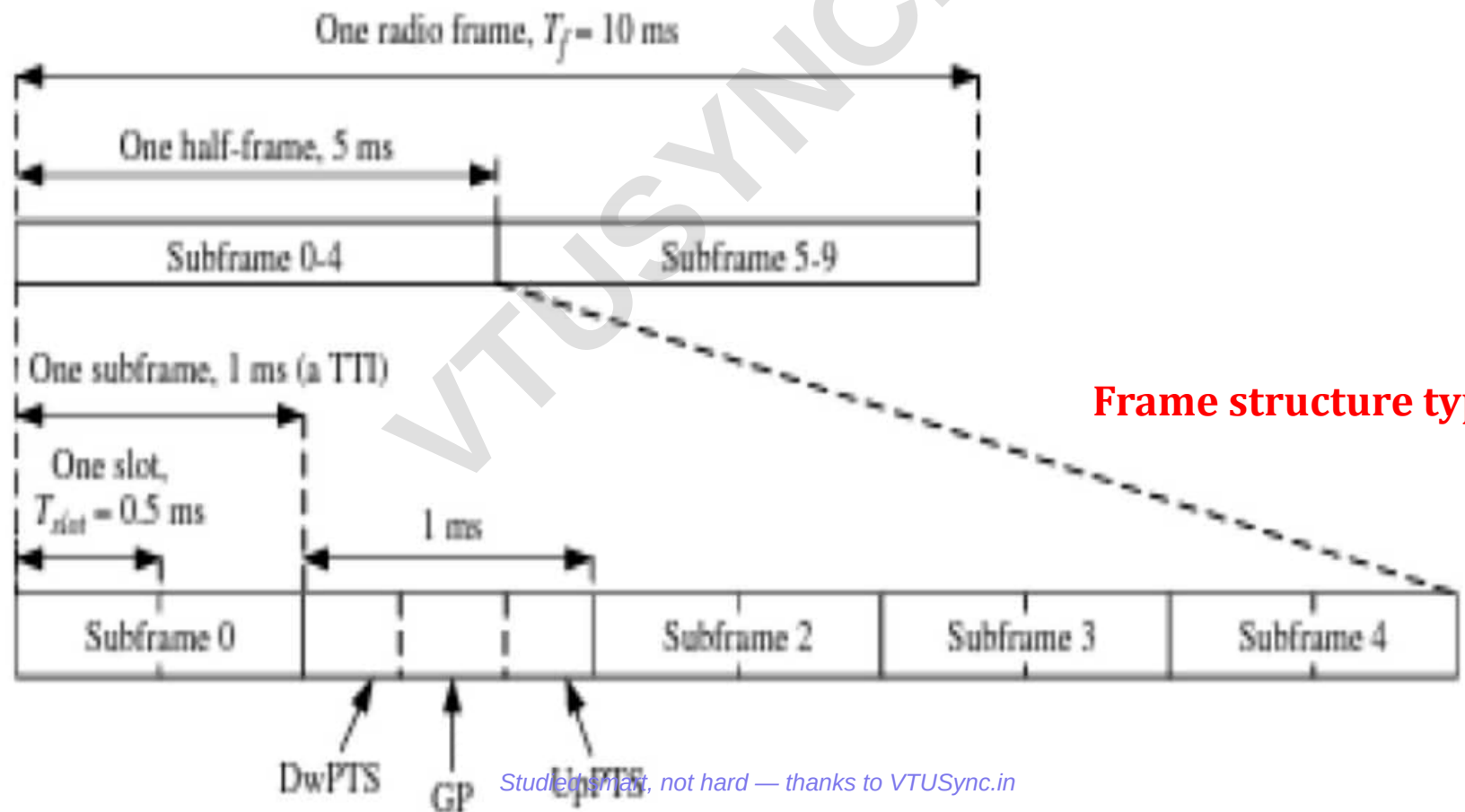
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Extended CP

Cyclic Prefix

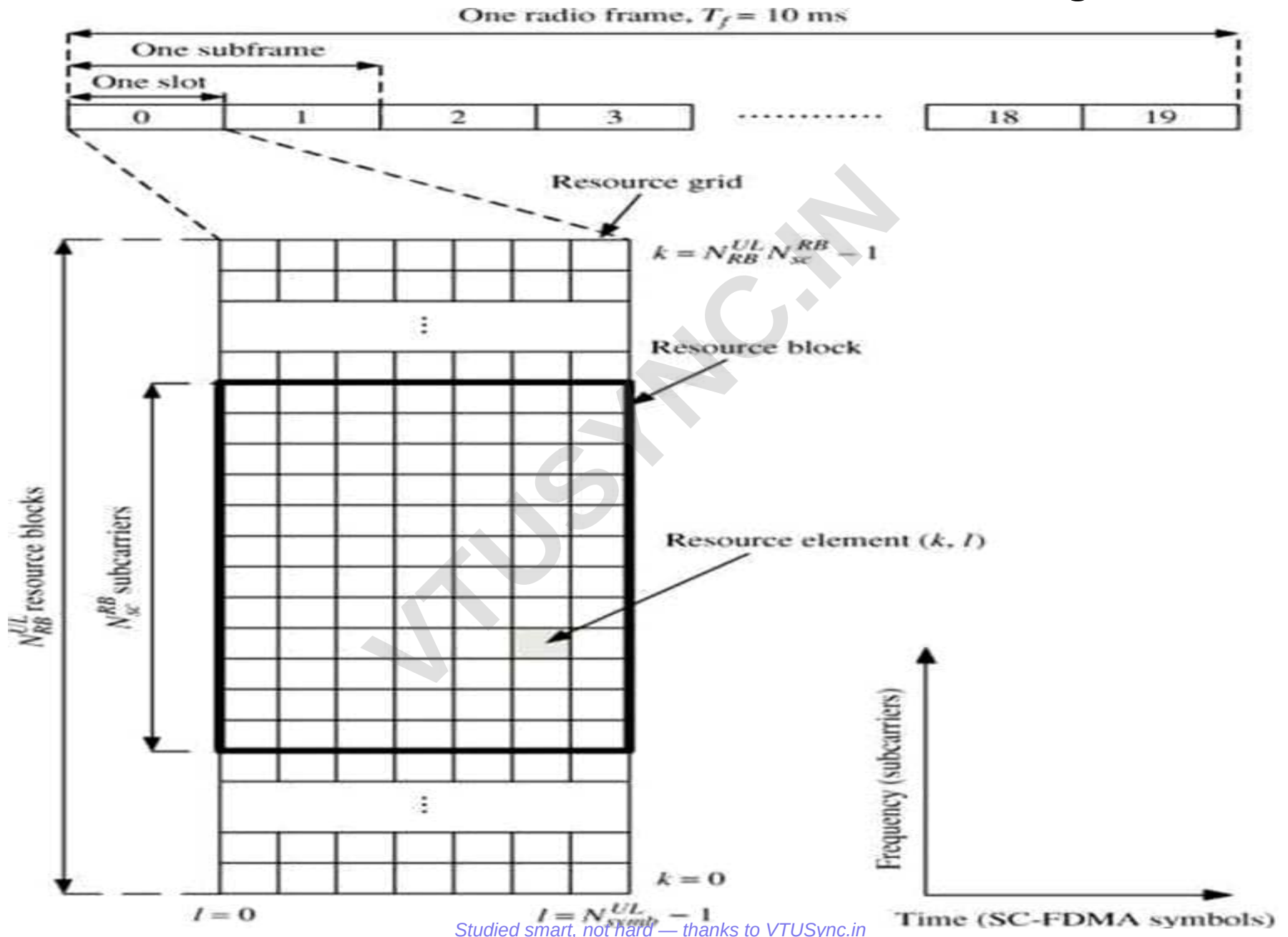


Frame structure type 2 consists of ten sub-frames, with one or two special sub frames including DwPTS, GP, and UpPTS fields, as shown. A CP is inserted prior to each SC-FDMA symbol. Each slot carries seven SC-FDMA symbols in the case of normal CP, and six SC-FDMA symbols in the case of extended CP.



Frame structure type 2

The structure of the downlink resource grid

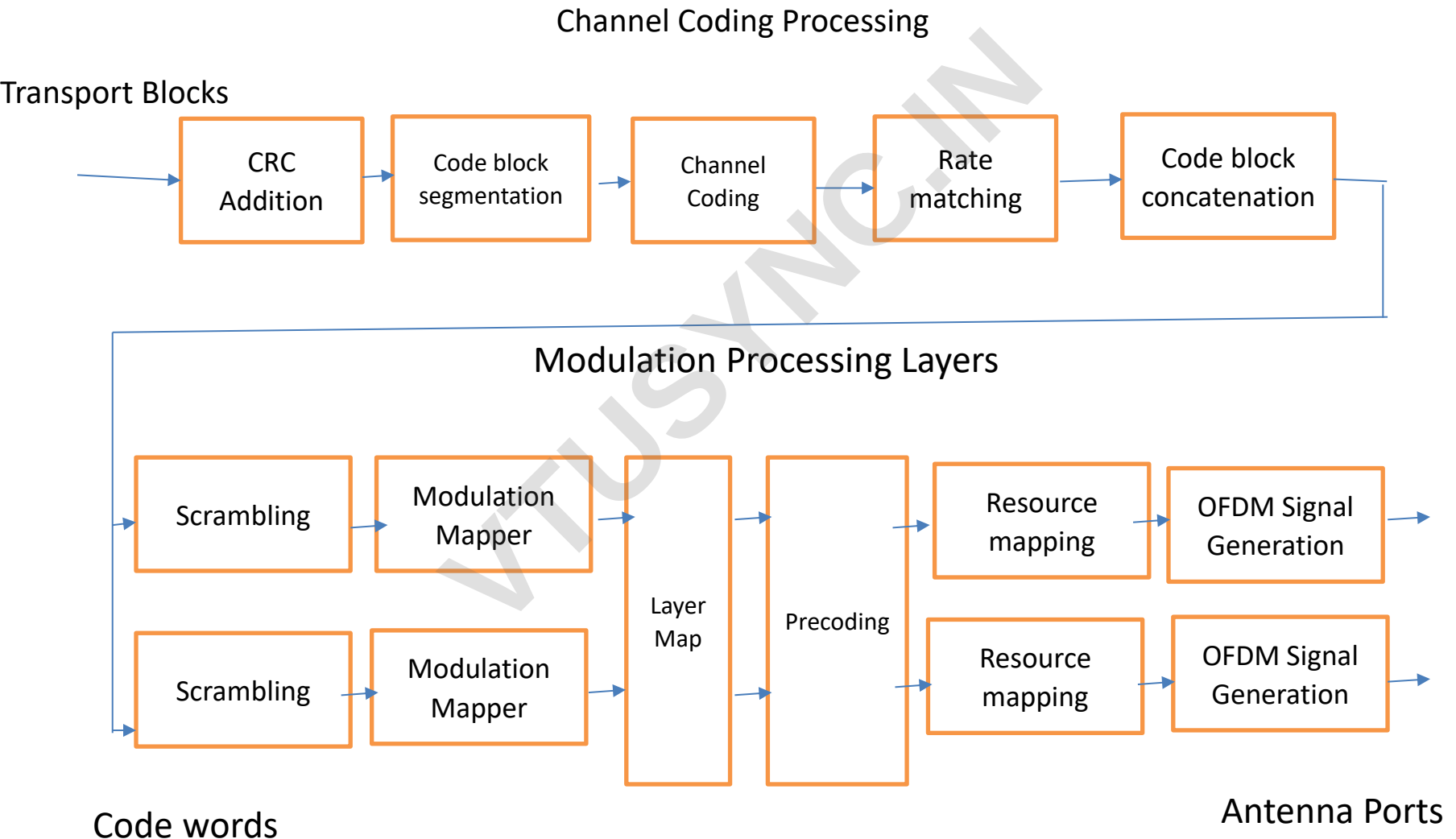


Uplink SC-FDMA Radio Resources

Resource Allocation:

- Resource allocation in the uplink is also performed at the eNode-B.
- Based on the channel quality measured on the uplink, **reference signals and the scheduling requests sent from Mobile**, the eNode-B assigns a **unique time frequency resource to a scheduled user**, which achieves orthogonal intra-cell transmission.
- Such intra-cell orthogonality in the uplink is preserved between mobile by using timing advance **such that the transport blocks of different mobiles are received synchronously at the eNode**

Downlink Transport Channel Processing

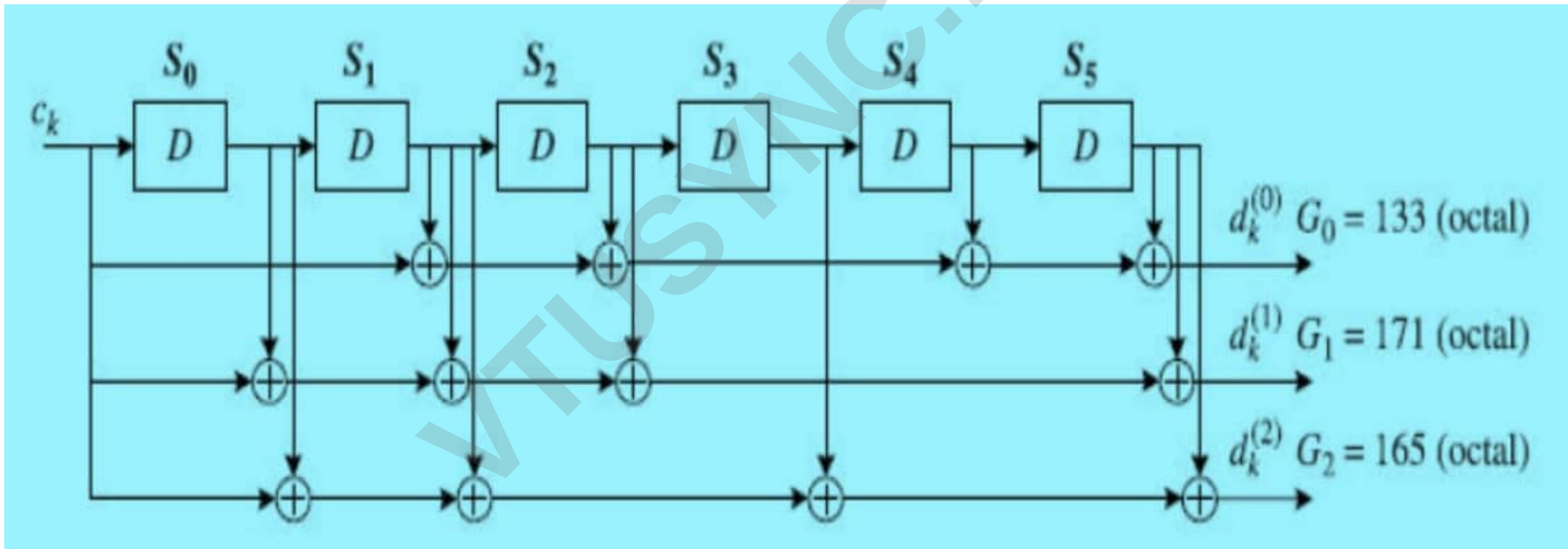


Channel Coding Processing

- channel coding procedures that are used for various data and control transport channels.
- These common aspects of channel coding are **applicable to both downlink and uplink transmissions.**
- Channel coding for the downlink is a combination of **error detection**, Error correction, rate matching, interleaving, and transport channel control information mapping onto physical channels.
- The error-control mechanism is coupled with the retransmission mechanism using what is called the Hybrid-ARQ (H-ARQ) protocol.
- This combined error-control and retransmission mechanism improves the link reliability

Channel Coding

Tail-Biting Convolutional Coding



Rate 1/3 tail-biting convolutional encoder.

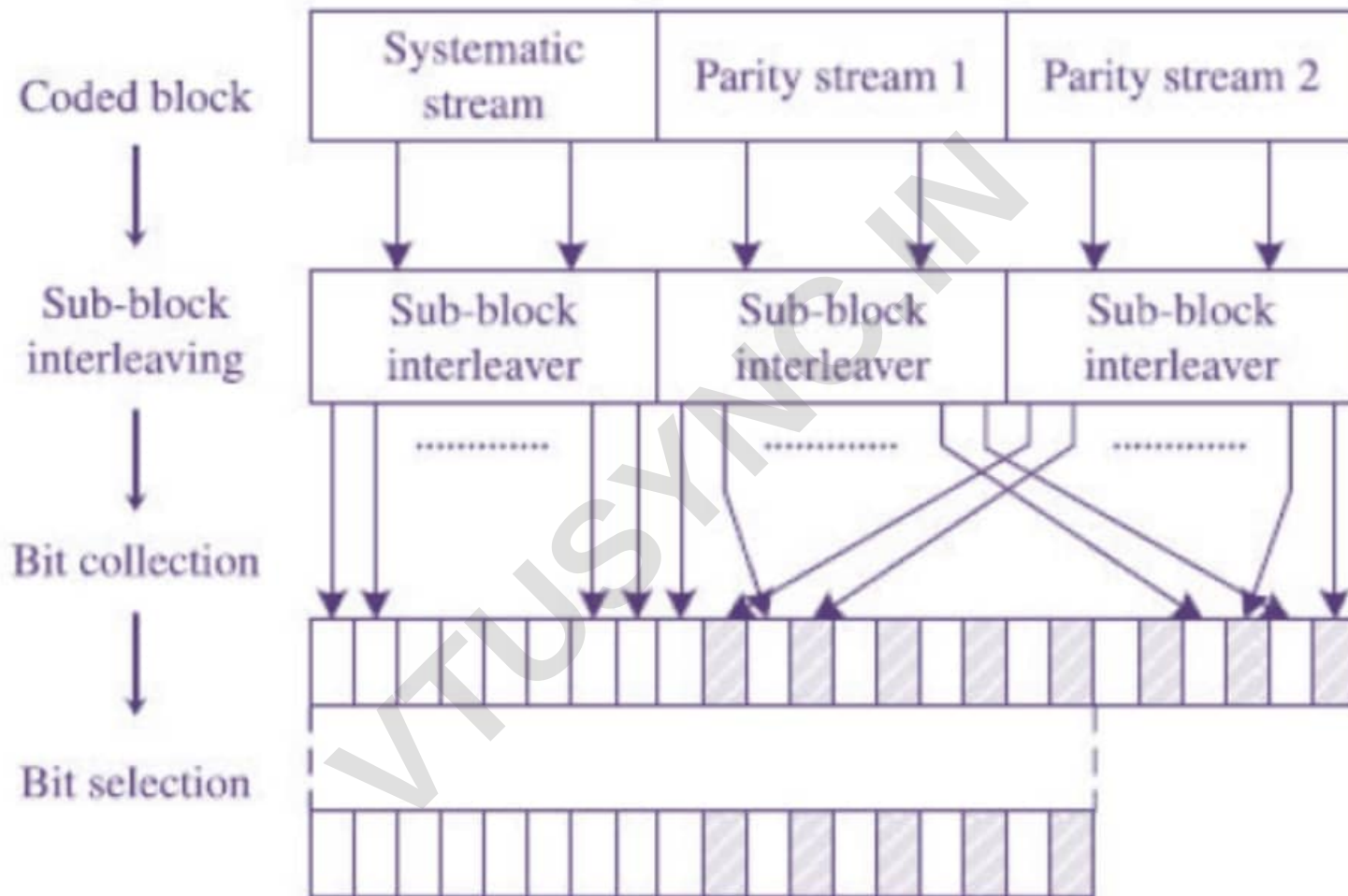


Figure 7.5 Rate matching for coded transport channels.

Code Block Concatenation

The code block concatenation consists of sequentially concatenating the rate matching outputs for different code blocks, forming the codeword input to the modulation processing. It is needed only for turbo coding when the number of code blocks is larger than one.

Scrambling

Before modulation, the codeword generated through channel coding processing is first scrambled by a bit-level scrambling sequence. The block of bits for codeword q is denoted as $b^{(q)}(0), \dots, b^{(q)}(M_b^{(q)} - 1)$, where $M_b^{(q)}$ is the number of bits transmitted in one subframe. The scrambling sequence $c^{(q)}$ is a pseudo-random sequence defined by a length-31 Gold sequence [3]. The scrambled bits are generated using a modulo 2 addition as:

$$\tilde{b}^{(q)}(i) = \left(b^{(q)}(i) + c^{(q)}(i) \right) \bmod 2, \quad i = 0, 1, \dots, M_b^{(q)} - 1.$$

Up to two codewords can be transmitted in the same subframe, so $q = 0$ if spatial multiplexing is not used or $q \in \{0, 1\}$ if spatial multiplexing is used.

Modulation Mapping

For each codeword q , the block of scrambled bits $\tilde{b}^{(q)}(0), \dots, \tilde{b}^{(q)}(M_b^{(q)} - 1)$ are modulated into a block of complex-valued modulation symbols $d^{(q)}(0), \dots, d^{(q)}(M_s^{(q)} - 1)$, where $M_s^{(q)}$ is the number of the modulation symbols in each codeword and depends on the modulation scheme. The relation between $M_s^{(q)}$ and $M_b^{(q)}$ is as follows:

$$M_s^{(q)} = \frac{M_b^{(q)}}{Q_m},$$

where Q_m is the number of bits in the modulation constellation, with $Q_m = 2$ for QPSK, $Q_m = 4$ for 16QAM, and $Q_m = 6$ for 64QAM.

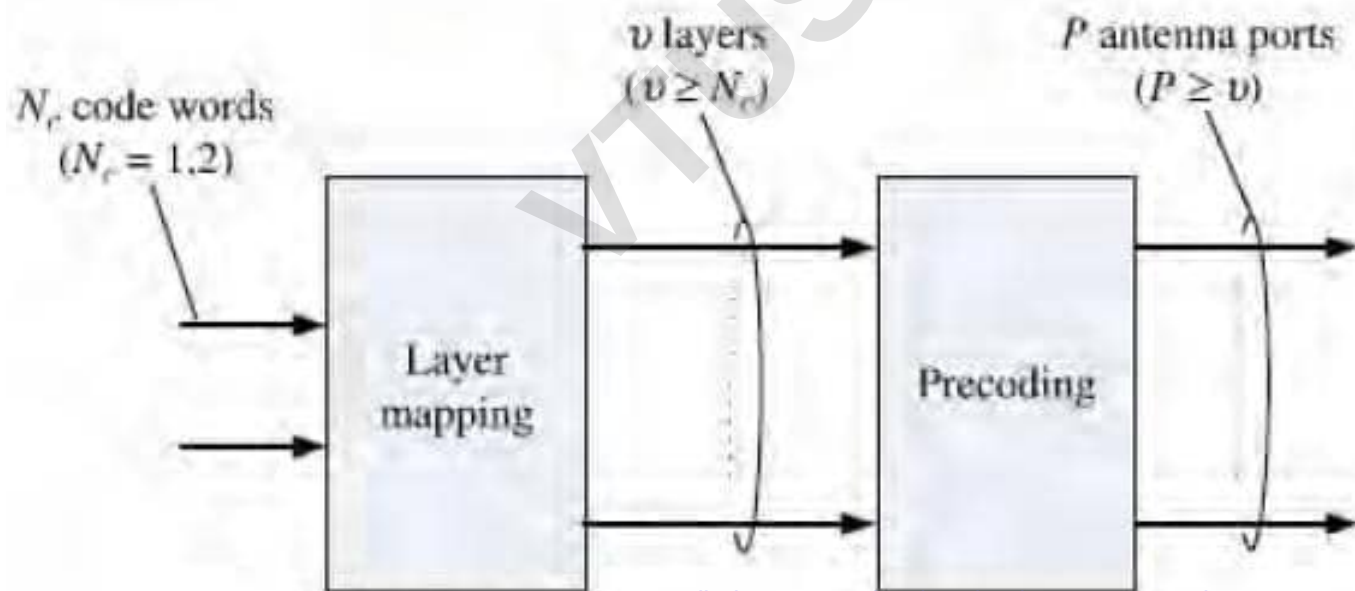
The supported data-modulation schemes in LTE include QPSK, 16QAM, and 64QAM, and BPSK is applied for the PHICH physical channel. Different physical channels employ different modulation schemes, listed in Table 7.3.

Physical Channel	Modulation Schemes
PDSCH	QPSK, 16QAM, 64QAM
PMCH	QPSK, 16QAM, 64QAM
PBCH	QPSK
PCFICH	QPSK
PDCCH	QPSK
PHICH	BPSK

Layer Mapping and Precoding

- **Codeword:** A codeword is defined as the output of each channel coding/rate matching stage associated with a single transport block coming from the MAC layer. For MIMO transmission with multiple codewords on different spatial channels, more efficient detectors with successive interference cancellation¹ can be used. In LTE, although up to four transmit/receive antennas are supported, the number of codewords is limited to two.

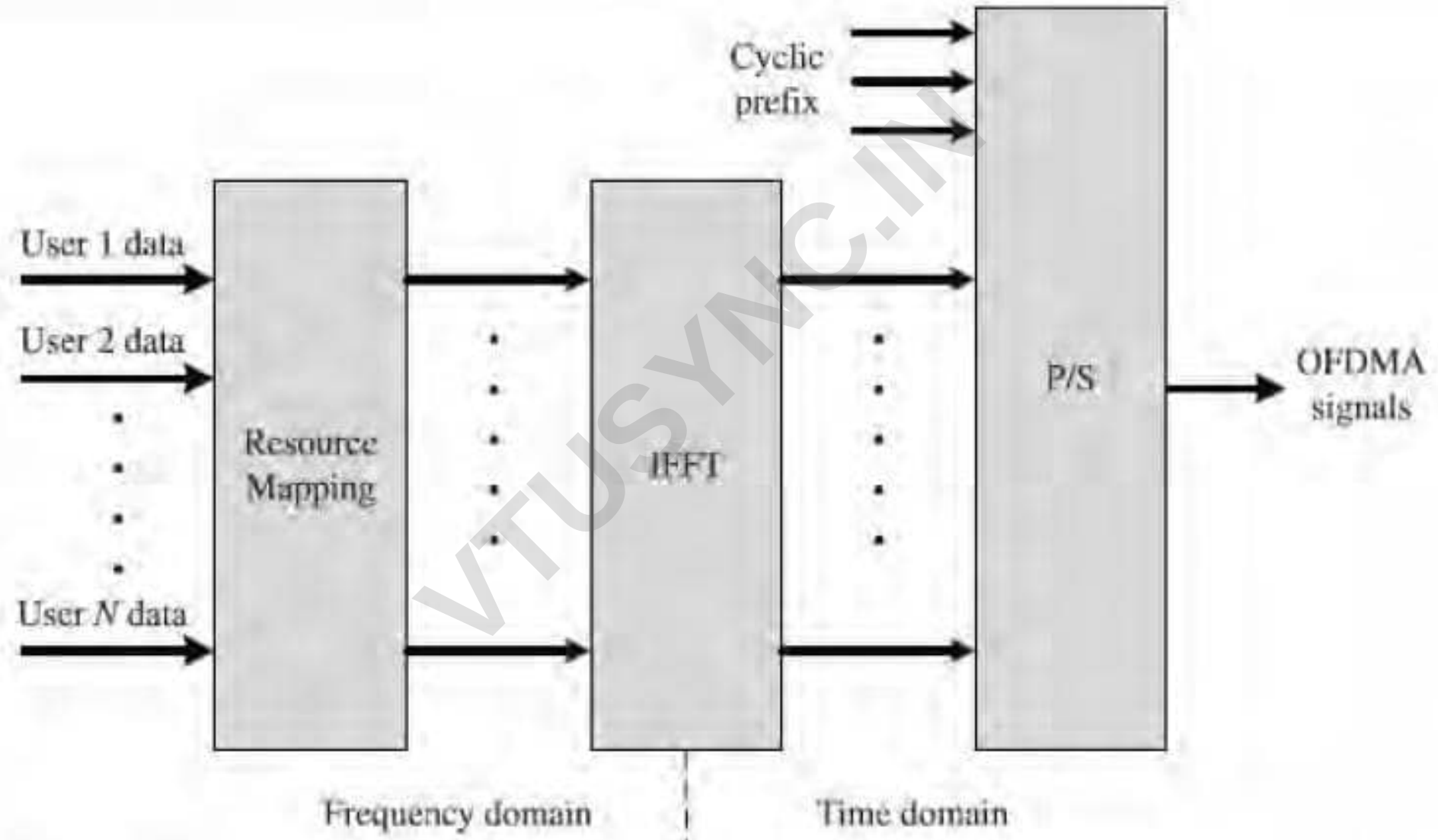
Layer: A layer corresponds to a data stream of the spatial multiplexing channel. Each codeword is mapped into one or multiple layers.



Antenna port: An antenna port is defined by its associated reference signal, which is a logical entity and may not correspond to an actual physical antenna. The number of transmit antenna ports at the eNode-B is sent to UEs through the PBCH channel, which can be 1, 2, or 4 in LTE. Antenna ports are divided into three groups:

- Antenna ports 0-3 are cell specific, which are used for downlink MIMO transmission.
- Antenna port 4 is MBSFN specific and is used for MBSFN transmission.
- Antenna port 5 is UE specific, which is used for beamforming to a single UE using all physical antennas.

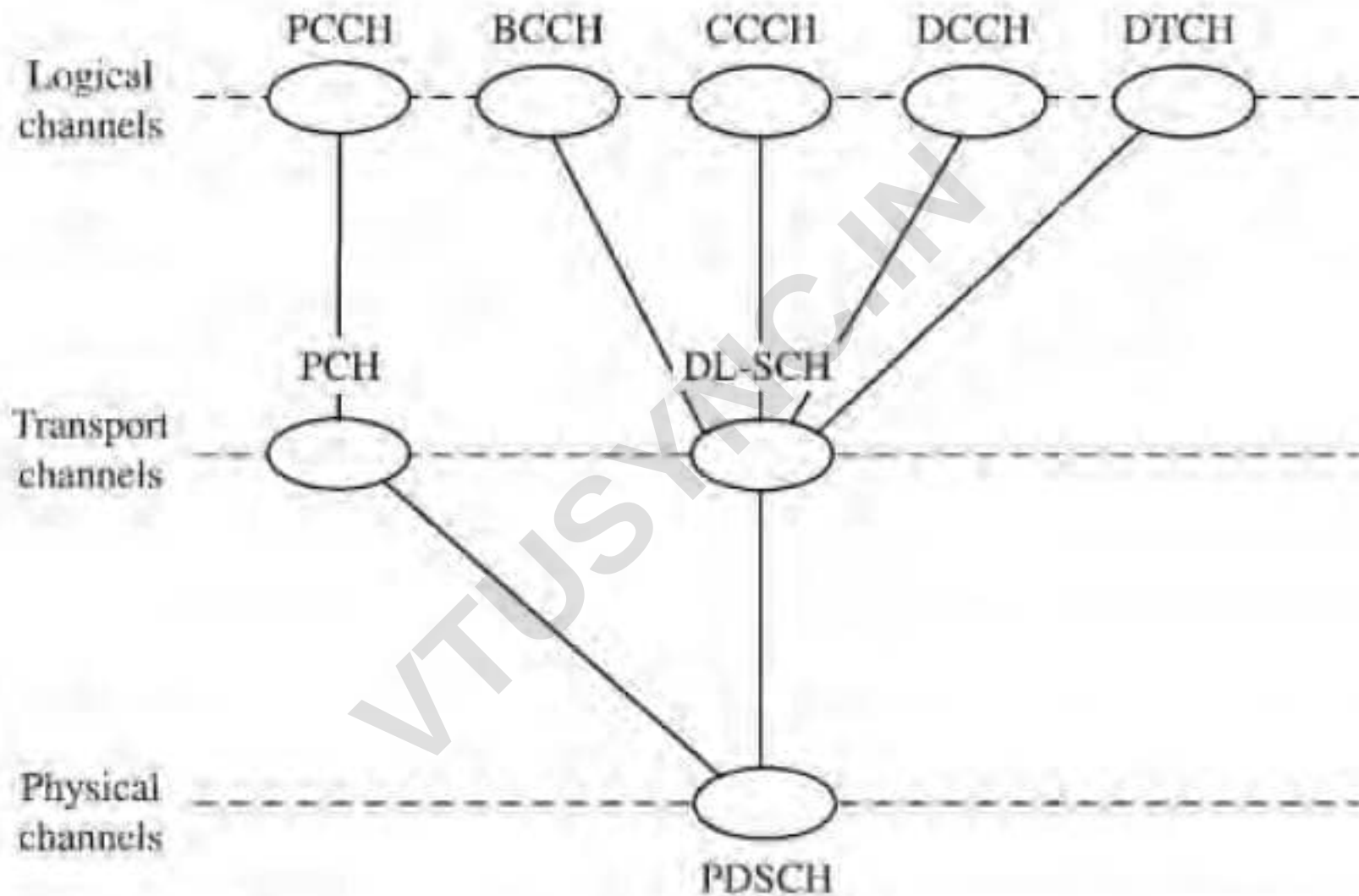
OFDM Baseband Signal Generation



Downlink Shared Channels

(DL-SCH):

- Transmitting the downlink data, including both control and traffic data, and thus it is associated with both logical control and logical traffic channels.
- able to **maximize the throughput** by allocating the resources to the optimum Users.
- The physical layer processing for specific transport channels including:
 - Downlink shared channel,
 - Downlink control channel,
 - Broadcast channel
 - Multicast channel.
- Although most of the transport channels implement the various aspects of transport channel processing such as **channel coding, rate matching symbol mapping, MIMO processing, and OFDM modulation**, the specifics of each step vary from one transport channel to the other



Channel mapping around the downlink shared channel.

Channel Encoding and Modulation

- Rate matching is used in order to achieve the effective channel coding rate that matches the payload capacity determined by the number of resource blocks allocated to the transport block of the given UE and the modulation scheme.
- The modulation scheme allowed for DL-SCH includes QPSK, 16QAM, and 64QAM and is chosen based on the Channel Quality Indicator (CQI) provided by the UE and various other parameters such as the size of the transport block.

Multiantenna Transmission

As the main channel for downlink traffic data transmission, the PDSCH supports all the MIMO modes specified in LTE.

There are seven different transmission modes defined for data transmission on the PDSCH channel:

Single-antenna port (port 0): One transport block is transmitted from a single physical antenna corresponding to internal port 0.

Transmit diversity: One transport block is transmitted from more than one physical antenna, that is, ports 0 and 1 if two physical antennas are used and ports 0, 1, 2, and 3 if four physical antennas are used.

Open-loop (OL) spatial multiplexing: One or two transport blocks are transmitted from two or four physical antennas. In this case, pre defined precoder matrices are used based on the Rank Indicator (RI) feedback(How well multiple antenna works). Precoding is fixed.(precoding: to support multi layer txn in MIMO)

Closed-loop (CL) spatial multiplexing: One or two transport blocks are transmitted from two or four physical antennas. The precoding in this case is adapted based on the Precoding Matrix Indicator (PMI) feedback from the UE.

Multiuser MIMO: Two Users are multiplexed onto two or four physical antennas with one transport block to each User. The rank-1 PMI feedback from each UE is used to create the overall precoding matrix.

Closed-loop rank-1 precoding: It is in special case of the CL spatial multiplexing with single-layer transmission, that is, a $P \times 1$ precoder is applied.

Single-antenna port (port 5): A single transport block is transmitted from two or more physical antennas.

- The eNode-B performs beamforming to a single user using all physical antennas.
- Unlike other modes, in this use the reference signal is also transmitted using the same beamforming vector that is used for the data symbols.
- Thus for this mode, the beamforming technique used at the eNode-3 is transparent to the UE, and the UE is able to decode the transport block with the help of this UE-specific reference signal.
- Beamforming can be used to improve the received signal power and/or reduce the interference signal power, which is Especially important for cell edge user's.

Downlink Control Channels

- Downlink control channels are carried over the Physical Downlink Control Channel (PDCCH) and they contain control information from the MAC layer, including downlink control information (DCI), Control Format Indicator (CFI), and H-ARQ Indicator (HI).
- Channel mapping between control information and physical channels in the downlink is shown in Figure .
- On the physical layer the PDCCH and the PDSCH are time multiplexed, such that the PDCCH is carried over the first few OFDM symbols of each sub frame and the PDSCH is carried over the rest of the OFDM symbols.
- The number of OFDM symbols allocated for PDCCH Can vary from one to four and is conveyed by the CFI.
- The CFI is carried on yet another control channel known as the Physical Control Format Indicator Channel (PCFICH)

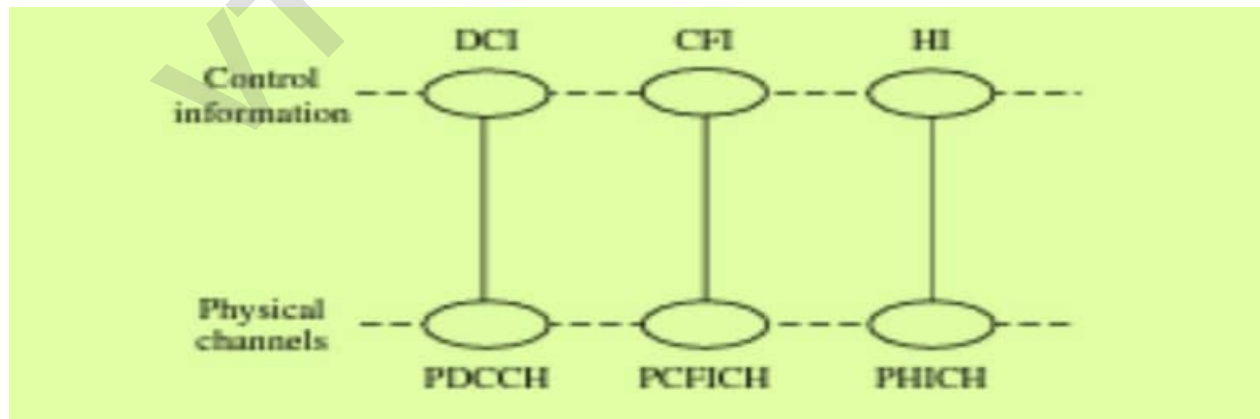


Figure: channel mapping for control information in the downlink

Downlink Control Information (DCI) Formats

- Among the control information in the downlink, **the DCI is the most important as it carries detailed control information for both downlink and uplink transmissions.**
- Therefore, in this section, we first describe different DCI formats. Then, the coding and modulation processing for each control channel is discussed, followed by the supported MIMO transmission.
- The DCI carries the downlink scheduling assignments, uplink scheduling
- **DCI format 0** carries uplink scheduling grants and necessary control information for uplink transmission.
- **DCI format 1/1A/1B/1C/1D** provides scheduling information for one code word transmission without spatial multiplexing.

H-ARQ Indicator (Hibrid automatic repeat request) :

- The control information HI is for H-ARQ acknowledgement in response to uplink transmission.
- As spatial multiplexing is not supported in the uplink in current specifications, only one information bit is required for H-ARQ acknowledgement.
- It has two values $HI = 1$ for 1 positive acknowledgment (ACK) and $HI = 0$ for a negative acknowledgment (NAK).
- A repetition code with rate $1/3$ is applied, which has two codewords: (0,0,0) and (1,1,1). Such a repetition is able to provide frequency diversity by mapping different code bits to different resource elements. BPSK modulation is applied.

Broadcast Channels

- Broadcast channels carry system information such as downlink system bandwidth, antenna configuration, and reference signal power.
- The UEs can get the necessary system information after the cell search (or synchronization) procedure.
- Due to the large size of the system information field, it is divided into two portions-Master Information Block (MIB) transmitted on the PBCH and System Information Blocks (SIB) transmitted on the PDSCH.
- The PBCH contains basic system parameters necessary to demodulate the PDSCH, which contains the remaining SIB. The transmission of the PBCH is characterized by a fixed pre-determined transport format and resource allocation, that is, there is no higher-layer control.

- Error detection is provided through a 16-bit CRC, and then the CRC parity bits are scrambled according to the eNode-B transmit antenna configuration with the scrambling sequence.
- The tail-biting convolution coding with rate $1/3$ is used, and the coded bits are rate matched to 1920 bits for the normal CP and to 1728 bits for the extended CP.
- The modulation scheme is QPSK, No H-ARQ is supported.
- For MIMO modes, PBCH supports single transmission and OL transmit diversity. Dynamic adaptation modulation and coding is not possible, due to the lack of channel quality feedback.

Multicast Channels

Multimedia Broadcast and Multicast Services (MBMS),

- supports multicast/broadcast services in 1 cellular system.
- It sends the same content information to all the UEs (broadcast) or to 1 given set of UEs (multicast). (delivering services such as mobile TV).
- In principle, the MBMS transmission can originate from a single base station or multiple base stations, but multicell transmission is preferred as large gains can be achieved through soft combining of transmissions from multiple base stations.
- For MBMS in the UTRA, to receive the same data information from multiple cells, the complexity at the receiver is high and there will be interference between signals from different base stations.
- One major design requirement for LTE is to provide enhanced support for the MBMS transmission, which is called *Enhanced MBMS (E-MBMS)* and is achieved through the so-called Single-Frequency Network (SFN) operation.

- With the OFDM-based transmission in the downlink, over-the-air combining of multicast, broadcast transmissions from multiple base stations is possible in LTE with an extended CP.
- The extended CP is used as the propagation delay from multiple cells and will typically be larger than the delay spread in a single cell.
- A longer CP can ensure that signals from different base stations still fall within the CP at the receiver, which avoids inter-symbol interference at the cost of a slight reduction in path data rate.
- In such *Multicast/Broadcast* Single Frequency Networks (MBSFNS), the same information is broadcast on the same radio resources from multiple synchronized neighbouring base stations to multiple UE..
- The SFN operation has already been used in the terrestrial Digital Video Broadcasting system (DVB-T).

There are two types of E-MBMS transmissions:

Single-cell transmission (non-MBSFN operation): The MBMS service (MTCH and MCCH) is transmitted on the MCH, and combining of MBMS transmission from multiple cells is not supported.

Multicell transmission (MBSFN operation): The MBMS service (MTCH and MCCH) is transmitted synchronously on the MCH, and combining is supported with the SFN operation.

The MBSFN and DL-SCH transmission can be multiplexed in a time-division multiplexing (TDM) manner on 1 subcarrier basis, but cannot be transmitted within the same subframe.

Downlink Physical Signals

Downlink Reference Signals

- Downlink reference signals consist of known reference symbols that are intended for downlink channel estimation at the UE needed to perform coherent demodulation.
- The time and frequency intervals are mainly determined by the characteristics of the channels.

Three different types of downlink reference signals:

Cell-specific reference signals, MBSFN reference signals,
and UE-specific reference signals.

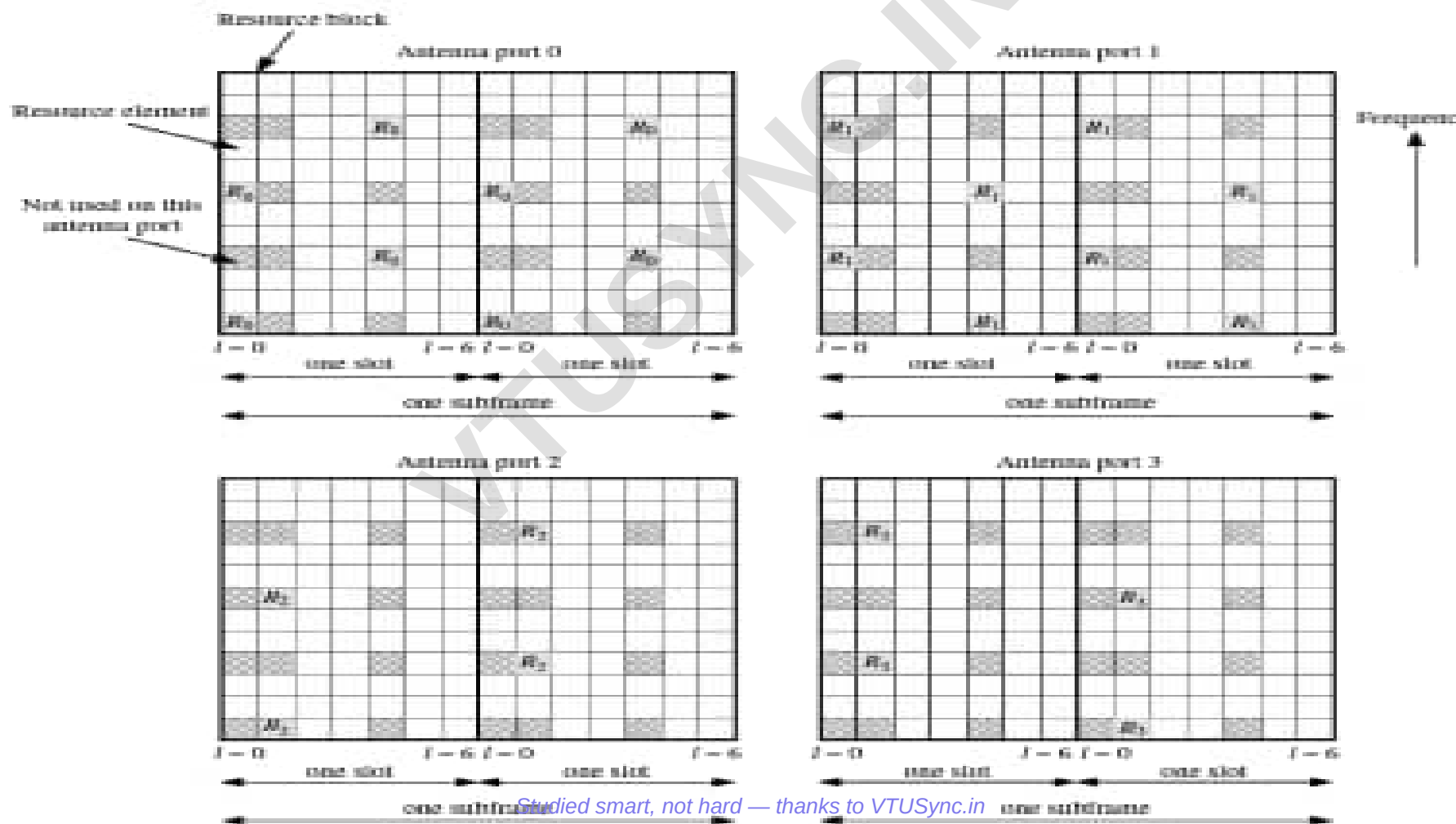
- The reference sequence is generated from a Pseudo-random sequence, with different initializations for different types of reference signals.
- There is one reference signal transmitted per downlink antenna port.

Cell-Specific Reference Signals:

- Cell-specific reference signals are transmitted in all downlink subframes in a cell supporting non-MBSFN transmission.
- In the subframe listed for transmission with MBSFN only the first two OFDM symbols can be used for cell-specific reference symbols. Cell specific reference signals are defined separately for antenna ports 0, 1, 2, and 3
- Therefore, in LTE a maximum of four antennas can be used while transmitting the cell specific reference signal.
- The cell specific reference signals are defined only for the normal subcarrier spacing of $\Delta f = 15\text{kHz}$.

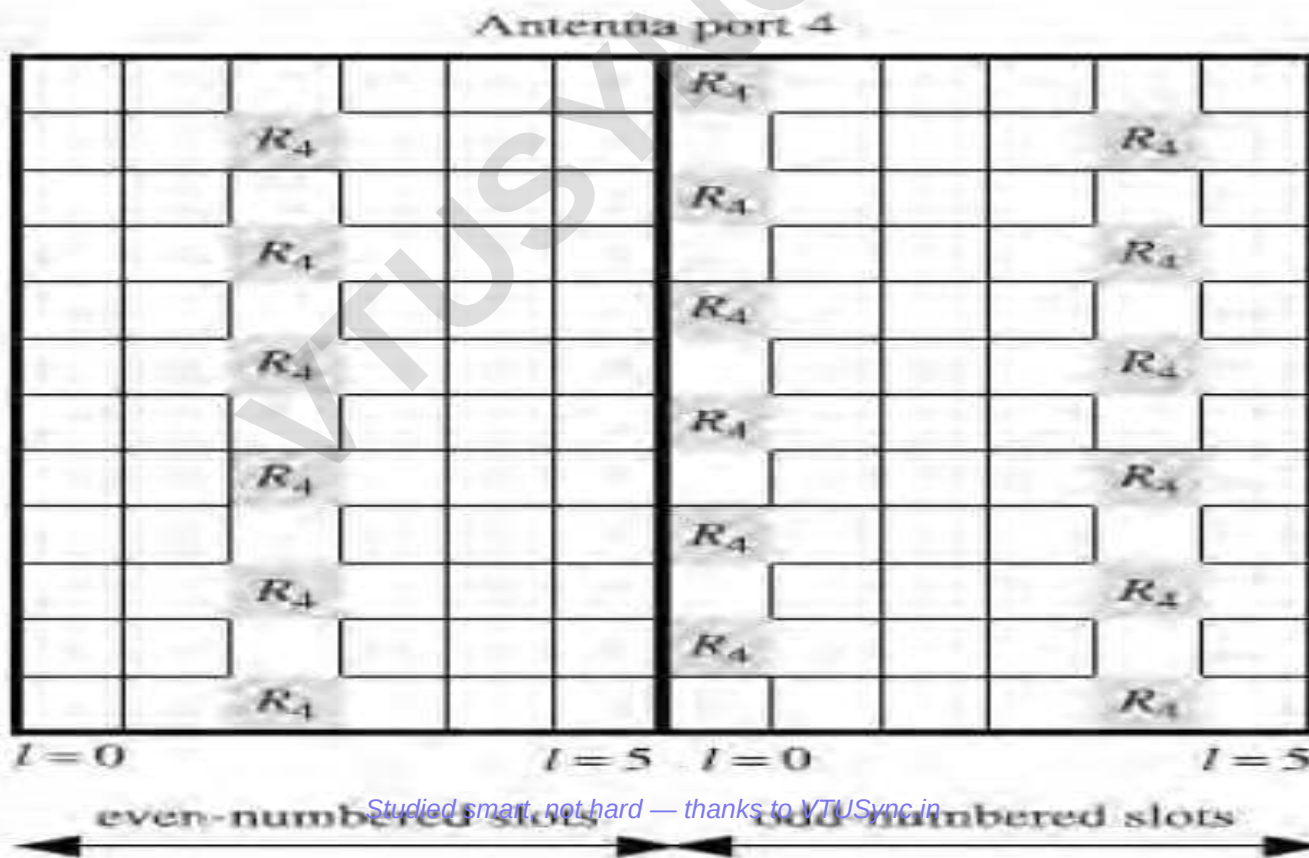
In the time domain, for the antenna port $\{0,1\}$, the reference symbols are inserted within the first and fifth OFDM symbols for the normal CP

For antenna port (2,3), the reference symbols are only inserted in the second OFDM symbol.

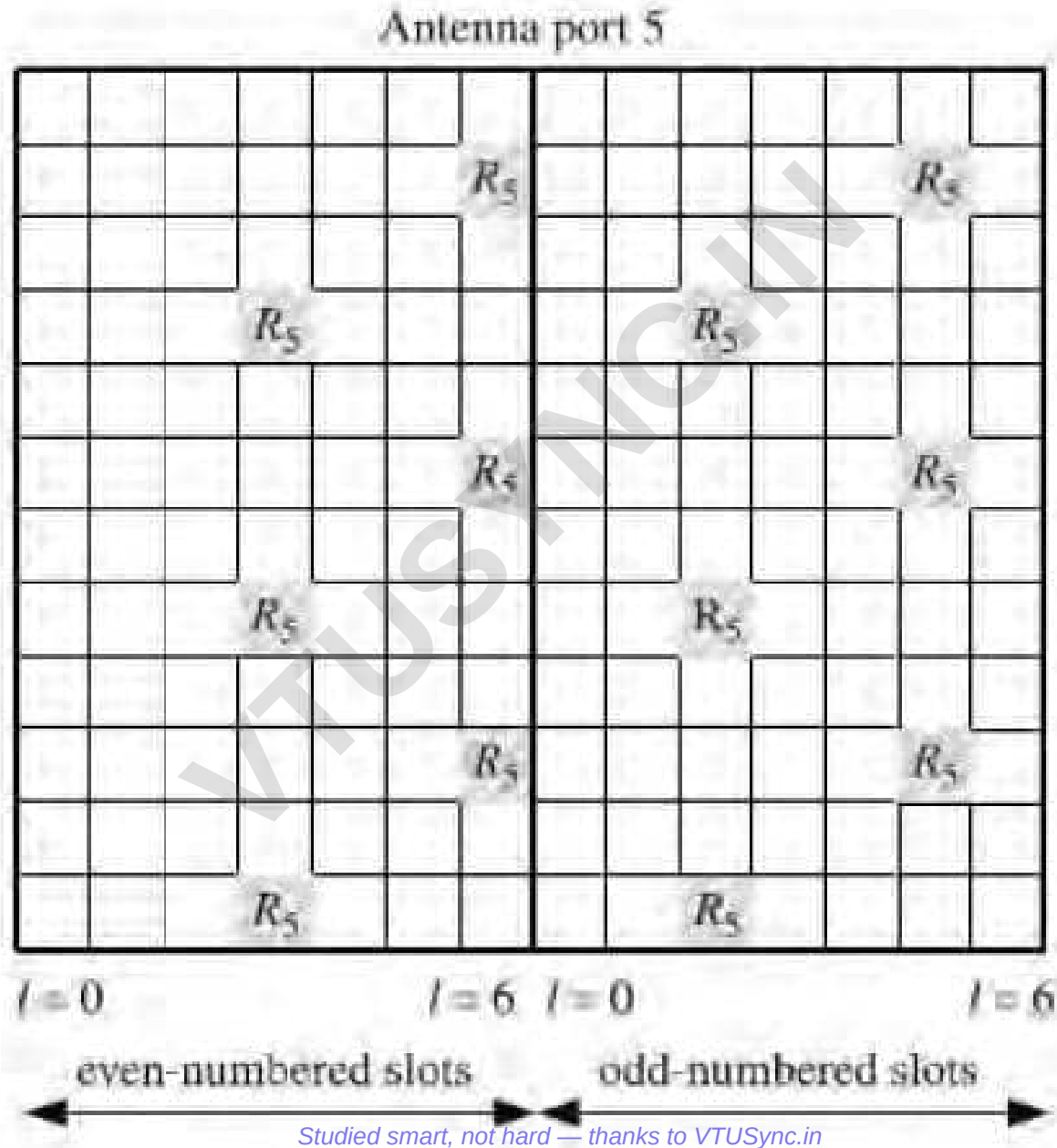


MBSFN Reference Signals

- MBSFN reference signals are only transmitted in sub frame allocated for MBSFN transmission, which is only defined for extended CP and transmitted on antenna port 4.
- In the time domain, for even-numbered slots, the reference symbols are inserted in the **third OFDM symbol** for $f = 15\text{kHz}$ and in the **last second OFDM symbol**.



UE-specific reference signals



H-ARQ in the Downlink (Hibrid automatic r repeat request)

- In a wireless network, due to the effects of channel fading and interference from neighbouring cells, it is nearly impossible to guarantee error-free transmission no matter how robust the channel coding is.
- Moreover, as the coding rate decreases, the transmission becomes more robust but at the same time power efficiency is lost, that is, a significant amount of power is used to transmit a few bits of information.
- An elegant approach to solve this problem is to use the H-ARQ protocol, which combines FEC and retransmission within a single framework.
- In case of LTE both Type I Chase Combining (CC) H-ARQ and Type II Incremental Redundancy (IR) H-ARQ schemes have been defined.

Thank You.....

End of Module -3